

```
# This is formatted as code
```

for loops

```
for i in range(1,5):  
    print(i)
```

```
1  
2  
3  
4
```

```
for i in range(7,2,-1):  
    print(i)
```

```
7  
6  
5  
4  
3
```

```
for i in range(1,10):  
    if (i==5):  
        print(i)
```

```
5
```

```
for i in range(1,21):  
    if (i%2==0):  
        print("even",i)  
    elif True:  
        if (i%1==0):  
            print("number odd",i)  
    else:  
        print(i)
```

```
number odd 1  
even 2  
number odd 3  
even 4  
number odd 5  
even 6  
number odd 7  
even 8  
number odd 9  
even 10  
number odd 11
```

```
even 12
number odd 13
even 14
number odd 15
even 16
number odd 17
even 18
number odd 19
even 20
```

```
for i in range(57,19,-1):
    if (i%2==0 and i%5==0):
        print("divide",i)
    else:
        print("not divided",i)
```

```
not divided 57
not divided 56
not divided 55
not divided 54
not divided 53
not divided 52
not divided 51
divide 50
not divided 49
not divided 48
not divided 47
not divided 46
not divided 45
not divided 44
not divided 43
not divided 42
not divided 41
divide 40
not divided 39
not divided 38
not divided 37
not divided 36
not divided 35
not divided 34
not divided 33
not divided 32
not divided 31
divide 30
not divided 29
not divided 28
not divided 27
not divided 26
not divided 25
not divided 24
not divided 23
not divided 22
not divided 21
divide 20
```

```
t=0
for i in range (1,11):
    t = t+i
    print(t)
```

```
1
3
6
10
15
21
28
36
45
55
```

```
# when a loop from 3,37 print when number is divided by 3 and 4
for i in range(3,37):
    if (i%3==0 and i%4==0):
        print("divided by both",i)
    else:
        print("not divide ",i)
```

```
not divide 3
not divide 4
not divide 5
not divide 6
not divide 7
not divide 8
not divide 9
not divide 10
not divide 11
divided by both 12
not divide 13
not divide 14
not divide 15
not divide 16
not divide 17
not divide 18
not divide 19
not divide 20
not divide 21
not divide 22
not divide 23
divided by both 24
not divide 25
not divide 26
not divide 27
not divide 28
not divide 29
not divide 30
not divide 31
```

```
not divide 32
not divide 33
not divide 34
not divide 35
divided by both 36
```

```
# when a loop from 5,40 and print only odd number
for i in range (5,40):
    if (i%2==0):
        print("even",i)
    else:
        print("odd",i)
```

```
odd 5
even 6
odd 7
even 8
odd 9
even 10
odd 11
even 12
odd 13
even 14
odd 15
even 16
odd 17
even 18
odd 19
even 20
odd 21
even 22
odd 23
even 24
odd 25
even 26
odd 27
even 28
odd 29
even 30
odd 31
even 32
odd 33
even 34
odd 35
even 36
odd 37
even 38
odd 39
```

```
# when a loop 97,48 and print sum all the even numbers
s = 0
for i in range (97, 48, -1):
    if i %2 == 0:
```

```
s = s+i  
print(s)
```

```
96  
190  
282  
372  
460  
546  
630  
712  
792  
870  
946  
1020  
1092  
1162  
1230  
1296  
1360  
1422  
1482  
1540  
1596  
1650  
1702  
1752
```

```
data = "regex"  
for char in data:  
    print (char)
```

```
r  
e  
g  
e  
x
```

```
data = "divya "          # positive indexing  
for char in range(0,5):  
    print(data[2])
```

```
v  
v  
v  
v  
v
```

```
data = "divya "          # negative indexing  
for char in data:  
    print(data[-3])
```

```
y  
y  
y  
y  
y  
y  
y
```

```
# form 2,40 and 3 multification  
for i in range (2,41):  
    if i%3==0:  
        print('divide',i)  
    else:  
        print("not divide",i)
```

```
not divide 2  
divide 3  
not divide 4  
not divide 5  
divide 6  
not divide 7  
not divide 8  
divide 9  
not divide 10  
not divide 11  
divide 12  
not divide 13  
not divide 14  
divide 15  
not divide 16  
not divide 17  
divide 18  
not divide 19  
not divide 20  
divide 21  
not divide 22  
not divide 23  
divide 24  
not divide 25  
not divide 26  
divide 27  
not divide 28  
not divide 29  
divide 30  
not divide 31  
not divide 32  
divide 33  
not divide 34  
not divide 35  
divide 36  
not divide 37  
not divide 38  
divide 39  
not divide 40
```

```
# 500,7 and multiply by 2, 3 ,8
for i in range (500,6,-1):
    if (i %2==0 and i % 3==0 and i%8==0):
        print ("divide",i)
    else:
        print("not divide",i)
```

```
not divide 500
not divide 499
not divide 498
not divide 497
not divide 496
not divide 495
not divide 494
not divide 493
not divide 492
not divide 491
not divide 490
not divide 489
not divide 488
not divide 487
not divide 486
not divide 485
not divide 484
not divide 483
not divide 482
not divide 481
divide 480
not divide 479
not divide 478
not divide 477
not divide 476
not divide 475
not divide 474
not divide 473
not divide 472
not divide 471
not divide 470
not divide 469
not divide 468
not divide 467
not divide 466
not divide 465
not divide 464
not divide 463
not divide 462
not divide 461
not divide 460
not divide 459
not divide 458
not divide 457
divide 456
not divide 455
```

```
not divide 454  
not divide 453  
not divide 452  
not divide 451  
not divide 450  
not divide 449  
not divide 448  
not divide 447  
not divide 446  
not divide 445  
not divide 444
```